

Elijah Shaw-Rutschman

Senior / Staff Software Engineer · Backend, Platform & Reliability · Open-Source Systems, Compilers & Embedded

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PROFESSIONAL SUMMARY

Software engineer with twenty years building production systems and apps in Python, JavaScript, Nim, C, and whatever else the job needed. Twelve years at StyleSeat, a B2C Python and Django marketplace serving ~2M consumers and 80K beauty professionals, owning the reliability, payments, and platform work the company depended on. Drove org-wide migrations under minimal guidance, including **Python 2-to-3 across ~1M lines in three months with no downtime**, done years before LLM coding assistants existed. Author, maintainer, and contributor to dozens of open-source libraries and tools other developers depend on, including the Nim compiler, Black, and Cython. My concurrency library lockfreequeues is part of the Nim compiler's regression suite. Titles along the way include Lead Django Developer and Senior Software Engineer II. Was on StyleSeat's formal Staff Engineer promotion track, holding their highest performance rating ("Exceptional"). Long tenure meant learning from the full multi-year lifecycle of product features and engineering decisions, which most engineers move on before seeing. Open to remote roles across the US.

CORE COMPETENCIES

Distributed Systems & Reliability

Python, Django & Celery

Platform & Release Engineering

AWS, Kubernetes, Terraform & Docker

Payments & Stripe Integration

Systems & Compiler Programming

Open-Source Maintenance

AI-Assisted Engineering

Mentoring & Technical Leadership

CI/CD, Testing & Quality

TypeScript & React.js

WORK EXPERIENCE

StyleSeat

Mar 2014 - Jun 2026

Senior Software Engineer II

Remote

- Reliability and failover: Architected the task-automation reliability layer, a Celery enqueue-failover path that captured broker-outage failures as durable dead-letter records, plus an S3-backed idempotent **recovery toolkit that replayed 351 tasks** lost during an outage. Owned a fault-tolerant Redis client library for graceful degradation when Redis was down.
- Rate limiting and DDoS defense: Built a Redis-backed API rate limiter (per-IP and per-user, sliding window, kill-switch) with a local-memory cache layer that **cut Redis calls by roughly 80-90%**, applied across auth, payment, and registration endpoints.
- Platform migrations at scale: Drove org-wide migrations under minimal guidance, including Python 2-to-3 across ~1M lines in three months with no downtime, plus pdm-to-uv, MySQL 5.7-to-8, and Segment-to-Braze.
- Platform, infrastructure and release engineering: Built and operated AWS infrastructure with Terraform, ECS, and Docker, and owned release tooling end to end - a custom CircleCI orb, deploy-gate and hotfix/cherry-pick workflows, and build-failure alerting - **cutting build times by roughly 40-50%**. Ran the systems delivered rather than handing them to a separate ops team.
- Payments: Owned payments as the team's domain expert across twelve years; major contributor to the Stripe billing and subscriptions build-out (promotion codes, plan backfills, coupon-stacking prevention, Stripe Portal, Pro Earnings), plus retry hardening on third-party Stripe calls and an automated fraud-detection system on Stripe Radar.
- AI code review: Designed StyleSeatBot, an org-wide automated code-review bot powered by Claude that became a required reviewer on every pull request, with adoption up to the CTO. Drove AI-assisted engineering adoption as a member of the AI-adoption tiger team.
- Senior ownership and mentorship: **The highest-volume code reviewer company-wide, year after year**, in a 75+ engineer organization; **mentored 10+ engineers** across product, platform, payments, and infrastructure. Earned the **highest performance rating ("Exceptional")** and was on the formal Staff Engineer promotion track. Role eliminated in a company-wide layoff (~1/3 of staff), June 2026.

Switchcam

Sep 2012 - Mar 2014

Software Engineer

San Francisco, CA / Chicago, IL

- Built audio-fingerprinting sync for crowd-sourced concert video, matching clips by their audio signature to align independently recorded footage on a shared timeline.
- Built the Python/Django backend at a social-video startup (data models, REST APIs) and the JavaScript front end for the product.

Fulton Works

Mar 2011 - Sep 2012

Lead Django Developer

Chicago, IL

- Led Python/Django development of client web applications at a development studio, including ZeroMQ message-passing between services, owning architecture and full-stack delivery across multiple client engagements.

Hagerty Consulting

Jun 2009 - Mar 2011

Software Engineer

Evanston, IL

- Built internal web applications and Python/Django data tooling supporting client engagements at an emergency-management and disaster-recovery consulting firm.

MSDSonline

Jun 2006 - Jun 2009

Software Engineer / IT Engineer

Chicago, IL

- Developed web application features for a SaaS chemical-safety and regulatory-compliance platform, moving into software engineering from an IT and internal-tooling role.

PROJECTS

Upstream contributions Contributor Merged into 40+ repos

Merged fixes into the Nim compiler (18.2k stars), covering its type system and C code generator, including cycle detection for recursive concepts and codegen for lifecycle hooks. Also merged upstream into Black (41.8k stars), asdf (25.5k), Homebrew Cask (22.2k), Cython (10.8k), typeshed (5.1k), nanomsg/NNG (4.7k), crosstool-ng (2.5k), Dropbear SSH (2.4k), RtAudio (1.8k), Zyre (948), and dozens more.

Python, Nim, C, C++, Compiler internals, Cross-compilation, Realtime audio, Messaging, Packaging

run-on-arch-action Top Contributor 749+ GitHub stars

GitHub Action running CI on non-x86 CPUs (ARM, aarch64, s390x, ppc64le, riscv64) via Docker and QEMU. Refactored it for maintainability and to lower the barrier to outside contributions; now the largest contributor to the project, ahead of the original author.

CI/CD, QEMU, Docker, Cross-architecture, GitHub Actions

pyng Maintainer 328+ GitHub stars

Maintainer and major contributor to the Python bindings for nanomsg next-generation (NNG), the messaging library succeeding nanomsg and ZeroMQ. Covers scalability protocols for messaging between processes on one machine or across a network.

Python, C, NNG, nanomsg, CFFI, Messaging protocols, Distributed systems

python-autopxd2 Owner 52+ GitHub stars

Primary maintainer of the most widely used fork of autopxd, which generates Cython .pxd declarations automatically from C headers. Recently refactored onto headerkit, a clang-based header-parsing library written to replace the original hand-rolled parser with a real libclang front end and pluggable backends.

Python, Cython, C, libclang, Parsing

lockfreequeues Author 50+ GitHub stars

Lock-free queues for Nim (SPSC, SPMC, MPSC, MPMC), and one of the community libraries in the Nim compiler's regression suite.

Nim, Lock-free concurrency, Systems

asdf-nim Author 27+ GitHub stars

Nim plugin for the asdf version manager, providing prebuilt-binary and source installs across platforms.

Nim, Shell, Packaging, Cross-platform, Developer tooling

ringbuf Author 22+ GitHub stars

Lock-free single-producer single-consumer ring buffer for Python and Cython, wrapping Boost.Lockfree so audio callbacks can hand data to Python without the GIL or allocation in the realtime path.

Python, Cython, C++, Lock-free concurrency, Realtime audio

build-farm Author 17+ GitHub stars

Cross-compilation toolchain distributing builds across Docker containers with distcc, targeting non-x86 architectures without native hardware.

Docker, distcc, Cross-compilation, Toolchains, CI/CD

axiomantic / spellbook Author AI Tooling

Multi-platform harness for AI coding assistants: skills, hooks, and a shared MCP server across Claude Code, OpenCode, Codex, and Gemini CLI. Most-starred project in the axiomantic org.
MCP, LLM, Developer experience, Python

axiomantic / momus Author AI Tooling

AI code review packaged as a GitHub Action, running automated review on pull requests.
LLM, AI code review, GitHub Actions, CI/CD, Python

tripwire (Nim) / pytest-tripwire (Python) Author

Test-mocking libraries that make verification mandatory: every external call is recorded and must be explicitly authorized and asserted, with unasserted interactions failing at teardown. 27 built-in integrations across HTTP, databases, caches, subprocesses and cloud services. Same design, two languages.
Nim, Python, pytest, Test mocking, Static analysis, Developer tooling

axiomantic / headerkit Author

Clang-based C/C++ header parsing toolkit in Python with pluggable backends for generating bindings and interop code. Extracted as a reusable library and now the parsing layer behind python-autopxd2.
Python, C/C++, libclang, Parsing, FFI, Developer tooling

nim-debra Author

Distributed epoch-based safe memory reclamation (DEBRA+) for lock-free data structures in Nim, reclaiming memory without blocking or pausing concurrent operations.
Nim, Lock-free concurrency, Memory management, Systems

axiomantic / komputers Author

Rust bindings for the Kompute GPU compute framework.
Rust, GPU compute, Vulkan, FFI

nim-tpestates Author

Compile-time state machine validation for Nim: invalid state transitions fail to compile rather than surfacing at runtime.
Nim, Type systems, Compile-time programming, Static verification

sonoscope Author Audio Tooling

Audio-QA harness for audio plugins, driving parameters to rendered wav and on to analysis JSON for automated regression testing of DSP output.
DSP, Audio analysis, Test automation, Python

PebbleOS port to the Bangle.js 2 Contributor Embedded

Porting PebbleOS to the Bangle.js 2 (nRF52840): touch input, bootloader, color display, and BLE, plus a Renode emulation for board bring-up without hardware.
Embedded C, Firmware, RTOS, nRF52840, BLE

Nord Modular G2 emulation Author & Contributor Emulation / DSP

Extending the gearmulator DSP56k emulator to emulate the Clavia Nord Modular G2 synthesizer. Spans an MCF5307 ColdFire core and ISP1181 device model in Nim, an MC68331 emulator in C++, firmware extraction and patch analysis tooling, and the dsp56300 emulator core.
C++, Nim, DSP56k, ColdFire, MC68331, Emulation, Reverse engineering, Firmware, Realtime audio

Plus dozens of other authored projects spanning compiler and interop tooling, audio and DSP libraries, developer tooling, emulators, GPU compute, and cross-platform build infrastructure.

EDUCATION

Bachelor of Science, Computer Science (Minor in Philosophy) **Northeastern Illinois University** 2008

SKILLS

Languages: Python, Go (Golang), Rust, Nim, C, C++, Objective-C, Cython, TypeScript, JavaScript, Lisp, SQL, Bash
Backend & Web: Django, Celery, FastAPI, REST API design, gunicorn, uvicorn, WSGI, ASGI, React (React.js), AngularJS (Angular.js), Node.js
Distributed Systems & Reliability: Distributed systems, High availability, Fault tolerance, Idempotency, Dead-letter queues, Failover, Rate limiting, Incident response, On-call rotation, ZeroMQ, nanomsg / NNG
Cloud, DevOps & Data: AWS, ECS, Kubernetes, Terraform, Infrastructure as code, DevOps, Docker, Jenkins, CircleCI, GitHub Actions, CI/CD, Release engineering, venv, virtualenv, pip, uv, Linux, MySQL, PostgreSQL, Elasticsearch, Redis
Testing & Quality: pytest, pylint, ruff, Test coverage, Static analysis, Code review
APIs, Auth & Payments: OAuth, OIDC, SSO, JWT, MFA, RBAC, Webhooks, Stripe billing, Stripe subscriptions, Stripe Radar, Fraud detection
Systems & Compilers: Lock-free concurrency, Memory management, Memory reclamation, Realtime audio / DSP, Audio fingerprinting, Compiler internals, Code generation, LLVM, libclang, Make, CMake, Build systems, Cross-compilation, Embedded C, Bare-metal, Firmware, RTOS
AI & LLM Tooling: AI-assisted engineering, Prompt engineering, LLM application development, AI agents, MCP (Model Context Protocol), FastMCP, Claude Code, AI code review